112159: porphyritic dacite, Royal Arthur

Location and sampling

LEONORA (SH 51-1) AMG Zone 51, 318200E 6854200N
Sampled on 1 August 1992

MIM Exploration drillhole WTB-86, 232–235 m.

Tectonic unit/relations

This sample is of a dacite from a belt of felsic extrusive rocks that hosts the Teutonic mine.

Petrographic description

This sample consists of a fine-grained quartzofeldspathic groundmass with abundant chlorite and microphenocrysts of plagioclase. Larger, euhedral phenocrysts of plagioclase, up to 4 mm across, and a few quartz phenocrysts, up to 2 mm across, together make up about 40% of the rock. Accessory carbonate and rutile are also present. The phenocrysts of calcic oligoclase are distinctive in that small blebs of quartz are common as inclusions in the rims of the grains. This appears to be a growth feature, and in extreme cases the phenocryst is internally vermicular. The groundmass contains a meshwork of small irregular veins of chlorite, possibly a late-stage cooling phenomenon of an extrusive rock. The rock is only lightly metamorphosed and no zircons were observed in the thin section described. GSWA Petrology Report Number 1615.

Zircon morphology

Abundant light brown euhedral to anhedral zircons, most between 150 and 300 µm long, and with magmatic zoning, were recovered from this sample.

Analytical details

Heavy minerals were isolated from about 1 kg of sample using conventional heavy-liquid and magnetic techniques.

This sample was analysed on 6 and 7 July 1994. The counter deadtime was 36 ns. Twelve analyses of the CZ-3 standard were obtained. The Pb^*/U calibration was stationary during the analysis session with a calibration error of 2.88 (1σ%).

Results

Forty-five analyses of 45 zircons were obtained. Results are given in Table 33 and shown on a concordia plot in Figure 34.

Interpretation

Forty-two samples have $^{207}Pb/^{206}Pb$ ratios belonging to single population with little excess scatter (chi-squared = 1.28) and indicating an interpreted extrusion age for this volcanic rock of 2692 ± 4 Ma. Analyses 31.1 and 42.1 have lower $^{207}Pb/^{206}Pb$ and fall statistically outside the main population. These analyses probably reflect some post-crystallization disturbance of these zircons. Analysis 30.1 has a significantly higher $^{207}Pb/^{206}Pb$ ratio, suggesting that this grain may be a xenocryst.

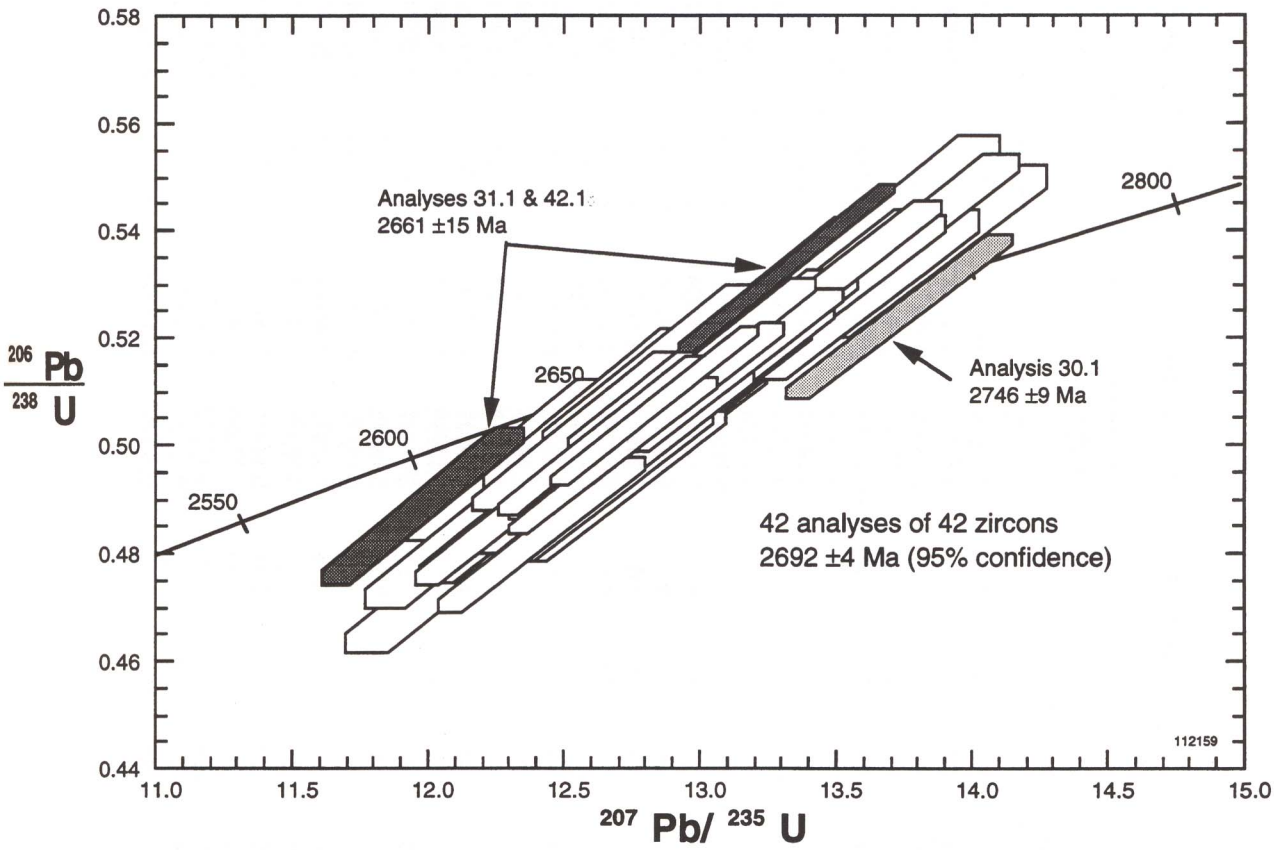

Figure 34. Concordia plot for sample 112159: porphyritic dacite, Royal Arthur

Table 33. Ion microprobe analytical results for sample 112159: porphyritic dacite, Royal Arthur

Grain .spot	*U (ppm)*	*Th (ppm)*	*Pb (ppm)*	*f206%*	$^{207}Pb/^{206}Pb$	*±1σ*	$^{208}Pb/^{206}Pb$	*±1σ*	$^{206}Pb/^{238}U$	*±1σ*	$^{207}Pb/^{235}U$	*±1σ*	*% concordance*	$^{207}Pb/^{206}Pb$ *Age*	*±1σ*
1.1	136	70	76	0.19	0.18654	0.00109	0.14366	0.00198	0.4830	0.0141	12.42	0.38	94	2 712	10
2.1	50	30	30	0.71	0.18505	0.00228	0.16202	0.00466	0.4944	0.0148	12.61	0.43	96	2 699	20
3.1	68	61	42	0.77	0.18006	0.00213	0.23854	0.00467	0.4968	0.0148	12.34	0.41	98	2 653	20
4.1	141	106	84	0.30	0.18527	0.00122	0.21154	0.00251	0.4946	0.0145	12.63	0.39	96	2 701	11
5.1	139	72	79	0.45	0.18415	0.00115	0.13901	0.00217	0.4941	0.0144	12.55	0.38	96	2 691	10
6.1	64	35	36	0.37	0.18434	0.00192	0.15622	0.00383	0.4757	0.0142	12.09	0.40	93	2 692	17
7.1	62	34	37	0.63	0.18093	0.00181	0.14347	0.00358	0.5057	0.0150	12.62	0.41	99	2 661	17
8.1	215	129	125	0.20	0.18289	0.00085	0.15823	0.00156	0.5014	0.0146	12.64	0.38	98	2 679	8
9.1	156	85	88	0.32	0.18324	0.00112	0.14910	0.00212	0.4881	0.0143	12.33	0.38	96	2 682	10
10.1	82	30	45	0.49	0.18342	0.00158	0.09476	0.00284	0.4889	0.0145	12.36	0.39	96	2 684	14
11.1	141	79	84	0.12	0.18549	0.00109	0.15326	0.00201	0.5135	0.0150	13.13	0.40	99	2 703	10
12.1	68	48	42	0.92	0.18570	0.00240	0.19940	0.00507	0.5004	0.0150	12.81	0.44	97	2 704	21
13.1	75	42	43	0.63	0.18609	0.00201	0.15900	0.00406	0.4926	0.0146	12.64	0.42	95	2 708	18
14.1	83	63	49	0.80	0.18231	0.00181	0.20629	0.00388	0.4838	0.0143	12.16	0.39	95	2 674	16
15.1	286	370	194	0.21	0.18381	0.00074	0.35801	0.00179	0.5067	0.0147	12.84	0.38	98	2 688	7
16.1	105	75	64	0.36	0.18382	0.00126	0.19134	0.00252	0.5056	0.0148	12.81	0.40	98	2 688	11
17.1	70	36	43	0.47	0.18697	0.00190	0.13980	0.00378	0.5272	0.0157	13.59	0.44	101	2 716	17
18.1	57	30	34	0.96	0.18311	0.00234	0.13900	0.00474	0.5021	0.0151	12.68	0.43	98	2 681	21
19.1	43	21	25	1.32	0.18293	0.00301	0.12978	0.00622	0.4971	0.0151	12.54	0.46	97	2 680	27
20.1	67	38	41	0.50	0.18550	0.00195	0.15580	0.00392	0.5147	0.0154	13.16	0.43	99	2 703	17
21.1	142	85	85	0.26	0.18628	0.00114	0.16353	0.00213	0.5102	0.0150	13.10	0.40	98	2 710	10
22.1	56	30	32	0.75	0.18646	0.00235	0.14637	0.00480	0.4928	0.0149	12.67	0.43	95	2 711	21
23.1	90	50	53	0.43	0.18597	0.00153	0.15005	0.00298	0.5071	0.0150	13.00	0.41	98	2 707	14
24.1	211	130	126	0.34	0.18475	0.00101	0.17024	0.00193	0.5072	0.0148	12.92	0.39	98	2 696	9
25.1	127	64	76	0.46	0.18399	0.00121	0.13865	0.00230	0.5165	0.0152	13.10	0.40	100	2 689	11
26.1	155	78	91	0.34	0.18245	0.00110	0.13515	0.00206	0.5134	0.0150	12.91	0.39	100	2 675	10
27.1	99	74	63	0.30	0.18586	0.00133	0.20409	0.00267	0.5262	0.0155	13.49	0.42	101	2 706	12
28.1	146	68	86	0.29	0.18537	0.00113	0.12696	0.00205	0.5157	0.0151	13.18	0.40	99	2 702	10
29.1	84	67	56	0.55	0.18340	0.00156	0.21362	0.00327	0.5408	0.0160	13.68	0.44	104	2 684	14
30.1	167	138	108	0.25	0.19047	0.00099	0.23464	0.00201	0.5233	0.0153	13.74	0.42	99	2 746	9

Table 33. (*continued*)

Grain .spot	*U (ppm)*	*Th (ppm)*	*Pb (ppm)*	*f206%*	$^{207}Pb/^{206}Pb$	$\pm 1\sigma$	$^{208}Pb/^{206}Pb$	$\pm 1\sigma$	$^{206}Pb/^{238}U$	$\pm 1\sigma$	$^{207}Pb/^{235}U$	$\pm 1\sigma$	*% concordance*	$^{207}Pb/^{206}Pb$ *Age*	$\pm 1\sigma$
31.1	125	58	69	0.36	0.17808	0.00128	0.12436	0.00233	0.4882	0.0143	11.99	0.37	97	2 635	12
32.1	131	64	75	0.19	0.18643	0.00111	0.13662	0.00195	0.4985	0.0146	12.81	0.39	96	2 711	10
33.1	145	111	88	0.44	0.18142	0.00120	0.20751	0.00248	0.5020	0.0147	12.56	0.39	98	2 666	11
34.1	43	23	26	0.67	0.18340	0.00244	0.13804	0.00491	0.5253	0.0160	13.28	0.46	101	2 684	22
35.1	90	45	55	0.36	0.18561	0.00140	0.13859	0.00257	0.5373	0.0159	13.75	0.43	103	2 704	12
36.1	61	39	38	0.64	0.18564	0.00209	0.17814	0.00428	0.5216	0.0157	13.35	0.45	100	2 704	19
37.1	134	87	81	0.38	0.18296	0.00127	0.17766	0.00254	0.5092	0.0149	12.85	0.40	99	2 680	12
38.1	77	36	48	0.61	0.18750	0.00191	0.13094	0.00378	0.5351	0.0159	13.83	0.45	102	2 720	17
39.1	83	57	51	0.44	0.18132	0.00167	0.18926	0.00341	0.5137	0.0153	12.84	0.41	100	2 665	15
40.1	183	116	108	0.27	0.18388	0.00106	0.17515	0.00207	0.5005	0.0146	12.69	0.39	97	2 688	10
41.1	252	320	167	0.24	0.18492	0.00090	0.34875	0.00215	0.4975	0.0145	12.69	0.38	97	2 697	8
42.1	275	306	189	0.08	0.18162	0.00067	0.31049	0.00152	0.5321	0.0155	13.33	0.40	103	2 668	6
43.1	144	139	96	0.23	0.18473	0.00111	0.26350	0.00239	0.5292	0.0155	13.48	0.41	102	2 696	10
44.1	162	97	97	0.28	0.18352	0.00108	0.16382	0.00202	0.5151	0.0151	13.03	0.40	100	2 685	10
45.1	88	42	53	0.45	0.18480	0.00149	0.13495	0.00282	0.5275	0.0156	13.44	0.43	101	2 696	13

STRATIGRAPHIC REFERENCE:

NELSON, D. R., 1997, Evolution of the Archaean granite–greenstone terranes of the Eastern Goldfields, Western Australia: SHRIMP U–Pb zircon constraints: Precambrian Research, v. 83, p. 57–81.

Recommended reference for this publication:

NELSON, D. R., 1995, 112159: porphyritic dacite, Royal Arthur; in Compilation of SHRIMP U–Pb zircon geochronology data, 1994: Western Australia Geological Survey, Record 1995/3, p. 128–131.
OR
NELSON, D. R., 1995, 112159: porphyritic dacite, Royal Arthur; Geochronology dataset 488; in Compilation of geochronology data, June 2006 update: Western Australia Geological Survey.

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